

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 620 Computational Geometry — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
David Mount — CMSC 754 Computational Geometry lecture notes (free PDF)	The primary source. A complete graduate course in one carefully written document, free, and better paced than most textbooks. Read alongside every module.
de Berg, Cheong, van Kreveld & Overmars — Computational Geometry: Algorithms and Applications	The standard text, and the source of the canonical treatments of sweep, Voronoi, and range searching. Library copy; the authors' course slides are free.
Jonathan Shewchuk — Adaptive Precision Floating-Point Arithmetic and Fast Robust Geometric Predicates (free)	The reference for Module 02 , with working code. The paper that made exact predicates practical rather than theoretical.
CGAL — the Computational Geometry Algorithms Library	The reference implementation of nearly everything in this course. Its documentation is itself a good secondary source, and its kernel design is the subject of Module 13.
Christer Ericson — Real-Time Collision Detection	The engine-facing companion. Where this course gives the algorithm, Ericson gives the shipping version. Library copy; the author's errata and notes are free.
TU Eindhoven — Computational Geometry course materials (free)	Slides and exercises from one of the groups that wrote the standard text.

Module-by-module map

Module 01 — Geometry a Machine Can Believe

Resource	What it covers for this module
Mount — CMSC 754 lecture notes, Lecture 1 (free PDF)	The framing of §1 and the orientation primitive, done carefully.
Shewchuk — Robust Adaptive Floating-Point Geometric Predicates (free)	The paper behind §2 and §3. Section 1 alone is the best statement of the problem in print, and Module 02 builds its machinery.
Kettner, Mehlhorn, Pion, Schirra & Yap — Classroom Examples of Robustness Problems in Geometric Computations (free)	Worked failures with figures: convex hulls that are not convex, produced by correct algorithms. Short, and it settles the matter for anyone still unconvinced.
Goldberg — What Every Computer Scientist Should Know About Floating-Point Arithmetic (free)	The cancellation analysis of §2, from first principles.

Module 02 — Exact and Adaptive Predicates

Resource	What it covers for this module
Shewchuk — Adaptive Precision Floating-Point Arithmetic and Fast Robust Geometric Predicates (free, with code)	The source for this entire module , including predicates.c. The error-bound derivations are in the paper and are worth following once.
Edelsbrunner & Mücke — Simulation of Simplicity (free)	The §3 technique, stated and proved by its authors.
Dekker; Knuth — two-sum and two-product	The expansion primitives of §2, with the correctness arguments.
CGAL — kernel and number type documentation (free)	How a production library packages exact predicates and inexact constructions, which is the Module 13 distinction.

Module 03 — Convex Hulls

Resource	What it covers for this module
Mount — CMSC 754, convex hull lectures (free PDF)	Monotone chain, gift wrapping, the lower bound, and Chan's algorithm, all done properly.
Barber, Dobkin & Huhdanpaa — The Quickhull Algorithm for Convex Hulls (free)	The paper and the qhull implementation, which is what most software actually calls in 3D and higher.
Chan — Optimal output-sensitive convex hull algorithms (free)	The $O(n \log h)$ result of §2, from its author. Short.
Ericson — Real-Time Collision Detection, chapters on convexity and bounding volumes	The §3 engine connection, in shipping terms.

Module 04 — Plane Sweep and Segment Intersection

Resource	What it covers for this module
de Berg et al. — Computational Geometry, chapter 2 (line segment intersection)	The canonical treatment of Bentley–Ottmann, including the degeneracy handling of §3.
Mount — CMSC 754, plane sweep lectures (free PDF)	The paradigm stated generally, then applied, which is the right order for it.
Bentley & Ottmann — Algorithms for reporting and counting geometric intersections	The original. Short, and worth reading for how directly the adjacency argument is made.
CGAL — 2D Arrangements package documentation (free)	A production implementation that handles every degeneracy in §3's table, and documents how.

Module 05 — Polygon Triangulation

Resource	What it covers for this module
de Berg et al. — Computational Geometry, chapter 3 (polygon triangulation)	The monotone decomposition of §3, with the five vertex types and the helper mechanism done precisely.
Mount — CMSC 754, triangulation and art gallery lectures (free PDF)	Includes Fisk's proof and the comb lower bound.
Mapbox — earcut (free, MIT)	A production ear clipper with hole support , small enough to read in an afternoon. The best source for how the bridging of §4 is actually done.
Shewchuk — Triangle (free)	Quality constrained Delaunay meshing, and the reference for the Module 09 route.

Module 06 — Polygon Boolean Operations

Resource	What it covers for this module
Greiner & Hormann — Efficient Clipping of Arbitrary Polygons (free)	The 1998 original of §2. Read it, then read the 2019 extension before implementing.
Foster, Overfelt & Hormann — Clipping simple polygons with degenerate intersections (free)	The degeneracy handling the original lacks. This is the version to implement.
Angus Johnson — Clipper2 (free, Boost licence)	The integer-coordinate strategy of §4, shipped. Widely used, readable, and the best evidence for that row of the table.
Hobby — Practical segment intersection with finite precision output; and CGAL's snap rounding package (free)	Snap rounding, stated properly, with the displacement bounds.

Module 07 — Spatial Subdivision

Resource	What it covers for this module
de Berg et al. — Computational Geometry, chapters 5, 12, 14 (range searching, BSP, quadtrees)	The analyses behind §2 and §3, including the range-query bound and the quadtree balancing argument.
Samet — Foundations of Multidimensional and Metric Data Structures	The exhaustive reference for everything in §3. The author's course slides are free and are a good substitute.
Ericson — Real-Time Collision Detection, chapters 6–8	The engine-facing treatment — grids, hierarchies, loose octrees, and the practical choices of §4.
Fabian Giesen — Morton codes and Z-order (free)	The bit arithmetic behind §3's implicit addressing, clearly explained.

Module 08 — Voronoi Diagrams

Resource	What it covers for this module
de Berg et al. — Computational Geometry, chapter 7 (Voronoi diagrams)	Fortune's algorithm with the beach line and the circle-event handling of §2, done in full.
Mount — CMSC 754, Voronoi and lifting lectures (free PDF)	The lifting map of §3 , derived rather than asserted.
Aurenhammer — Voronoi Diagrams: A Survey of a Fundamental Geometric Data Structure	The variants of §4, comprehensively. Still the best single survey.
Du, Faber & Gunzburger — Centroidal Voronoi Tessellations (free)	Lloyd's algorithm and CVT, with the convergence theory.

Module 09 — Delaunay Triangulation and Meshing

Resource	What it covers for this module
Shewchuk — Triangle, and Delaunay Refinement Algorithms for Triangular Mesh Generation (free)	The reference for §3 , with working code. Shewchuk's writing on meshing is unusually clear.
de Berg et al. — Computational Geometry, chapter 9 (Delaunay triangulations)	Incremental construction, the flipping termination argument, and the randomisation analysis of §2.
Shewchuk — What Is a Good Linear Finite Element? (free)	Why angle quality matters , derived from the interpolation error rather than asserted. The best answer to 'why not just use any triangulation'.
Cheng, Dey & Shewchuk — Delaunay Mesh Generation	The three-dimensional material of §4, including slivers and the Schönhardt obstruction. Library copy.

Module 10 — Range Searching and Proximity

Resource	What it covers for this module
de Berg et al. — Computational Geometry, chapters 5 and 10 (range searching)	Range trees and fractional cascading, with the analyses of §1.
Mount — CMSC 754, range searching lectures (free PDF)	Clearer on the canonical-node decomposition than most treatments.
Malkov & Yashunin — Efficient and robust approximate nearest neighbor search using HNSW (free)	The §3 default method , from its authors.
Facebook Research — FAISS (free)	Production implementations of IVE, product quantisation, and HNSW, with benchmarks. The practical companion to §3.

Module 11 — Visibility and Motion Planning

Resource	What it covers for this module
LaValle — Planning Algorithms (free book)	The reference for §2 and §4 , free in full. Configuration space, PRM, and RRT from the author of RRT.
de Berg et al. — Computational Geometry, chapters 13 and 15 (robot motion planning, visibility graphs)	Minkowski sums and visibility graphs, with the complexity analyses.
Mikko Mononen — Recast and Detour (free)	The navmesh pipeline of §3, implemented. The standard open-source navigation toolkit; the source is the documentation.
Amit Patel — Red Blob Games, pathfinding and visibility (free)	Interactive explanations of A*, the funnel algorithm, and 2D visibility. Unusually good for building intuition before reading the formal treatments.

Module 12 — Geometry in Three Dimensions

Resource	What it covers for this module
Ericson — Real-Time Collision Detection	The reference for this whole module. GJK, SAT, bounding volumes, and the practical choices of §3. Library copy; the author's notes and errata are free.
Casey Muratori — Implementing GJK (free video)	The clearest explanation of §1 available , built up from the support-function idea rather than from the published pseudocode.
Mamou — V-HACD (free)	The approximate convex decomposition of §3, implemented and widely used.
Attene, Campen & Kobbelt — Polygon Mesh Repair: An Introduction (free)	The §4 defect taxonomy , done systematically, with what can and cannot be repaired.

Module 13 — Shipping Geometric Code

Resource	What it covers for this module
CGAL — kernel documentation and the robustness manual (free)	The §1 strategy table, as a working library's design. The kernel-choice discussion is the best practical writing on this subject.
Kettner et al. — Classroom Examples of Robustness Problems (free)	Worth re-reading now that you can fix everything in it.
Chen, Cheung, Xie, Zhang & Zhang — Metamorphic testing: a review (free)	The §2 technique, generalised well beyond geometry.
Shewchuk — Lecture notes on geometric robustness (free)	The practitioner's summary, from the person who did most to make this tractable.